
Corporate Credit Card Misuse Detection Using Machine Learning and ETL Pipeline

Project Update 2 — Data & Formal Definitions

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Abstract

Corporate credit card misuse is a significant financial and compliance challenge in enterprise settings. This paper presents an end-to-end machine learning system for detecting fraudulent corporate card transactions, combining a full ETL pipeline (PostgreSQL, Apache Airflow) with a five-model comparison (Logistic Regression, LinearSVC, Random Forest, XGBoost, LightGBM). We use the ULB Credit Card Fraud Detection dataset (284,807 transactions, 0.17% fraud) and address severe class imbalance using SMOTE. Results are served through a FastAPI backend and React dashboard for production-style deployment.

1. Introduction and Problem Statement

Corporate credit card misuse is a significant financial and compliance challenge. Employees issued corporate cards for legitimate business expenses may use them for unauthorized personal purchases, resulting in financial losses and policy violations. Manual review of thousands of transactions is time-consuming and not scalable.

Formally, the learning problem is defined as: given a transaction record $x \in R^{30}$ consisting of 28 anonymized behavioral features $V_1, \dots, V_{28}$, a normalized transaction amount, and hour of day, predict a binary label $y \in \{0, 1\}$ where $y = 1$ indicates misuse and $y = 0$ indicates legitimate spending. The dataset used is the Credit Card Fraud Detection dataset (ULB, Kaggle) containing 284,807 transactions with 99.83% legitimate vs. 0.17% fraudulent — a severely imbalanced classification problem.

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2. Motivation

Financial fraud detection is one of the most practically important applications of machine learning. Organizations lose significant revenue to fraud annually, with corporate card misuse among the most common types. Automated detection systems reduce investigation time and improve accuracy compared to rule-based systems. Multiple studies have shown that models like Random Forest and Logistic Regression can effectively detect fraudulent transactions (Wang et al., 2023; Hande et al., 2019). The core challenge is severe class imbalance, which requires techniques like SMOTE oversampling and data augmentation (Khalid et al., 2024; Strelcenia and Prakoonwit, 2023).

3. Related Work

Logistic Regression is a common baseline for fraud classification because it is simple and interpretable. Wang et al. compared Logistic Regression and Random Forest on simulated transaction data and found Random Forest performs better (Wang et al., 2023). Sopiyan et al. also compared Logistic Regression with Random Forest and Gradient Boosting using SMOTE and found ensemble methods outperform the logistic baseline (Sopiyan et al., 2022).

Support Vector Machines have been used in fraud detection for finding decision boundaries in high-dimensional spaces. Xia et al. applied SVM with GridSearchCV tuning to the Kaggle credit card dataset, optimizing AUC and F1-score (Xia, 2022). Aftab et al. used SVM alongside Random Forest and Logistic Regression and found ensemble methods outperform single classifiers (Aftab et al., 2023).

Random Forest is one of the most effective models for fraud detection because its ensemble voting mechanism naturally handles class imbalance. Hande et al. compared Logistic Regression, Decision Tree, and Random Forest and found Random Forest performs best (Hande et al., 2019). Aburbeian and Ashqar combined Random Forest with SMOTE and achieved strong results on imbalanced credit card data (Aburbeian and Ashqar, 2023).

Gradient Boosting methods are the current state of the

art for tabular fraud detection. XGBoost builds trees sequentially, each correcting the residual errors of the previous. LightGBM uses leaf-wise tree growth, which makes it faster on large datasets. Sopiyan et al. found Gradient Boosting competitive with Random Forest on credit card fraud data (Sopiyan et al., 2022).

Class Imbalance Handling: SMOTE creates synthetic minority class samples by interpolating between existing ones. Khalid et al. tested ensemble methods with SMOTE on European credit card transactions and showed that re-sampling improves minority class detection (Khalid et al., 2024). Tran and Dang compared SMOTE and ADASYN for balancing credit card fraud data before training Random Forest and Logistic Regression (Tran and Dang, 2021). Strelcenia and Prakoonwit proposed K-CGAN as an alternative to SMOTE (Strelcenia and Prakoonwit, 2023).

Production ML Pipelines: Most prior studies focus only on model accuracy without considering deployment. ETL pipelines that separate ingestion, transformation, and loading ensure reproducibility and scalability. This project adopts that principle using PostgreSQL and Apache Airflow.

4. How This Approach Differs

Most prior work focuses only on model performance in isolation. This project combines a complete data engineering layer (ETL pipeline, PostgreSQL, Apache Airflow) with a five-model comparison (Logistic Regression, LinearSVC, Random Forest, XGBoost, LightGBM), a REST API backend (FastAPI), and an interactive React dashboard. This end-to-end production-style system addresses real-world deployment considerations that are typically missing from academic fraud detection papers.

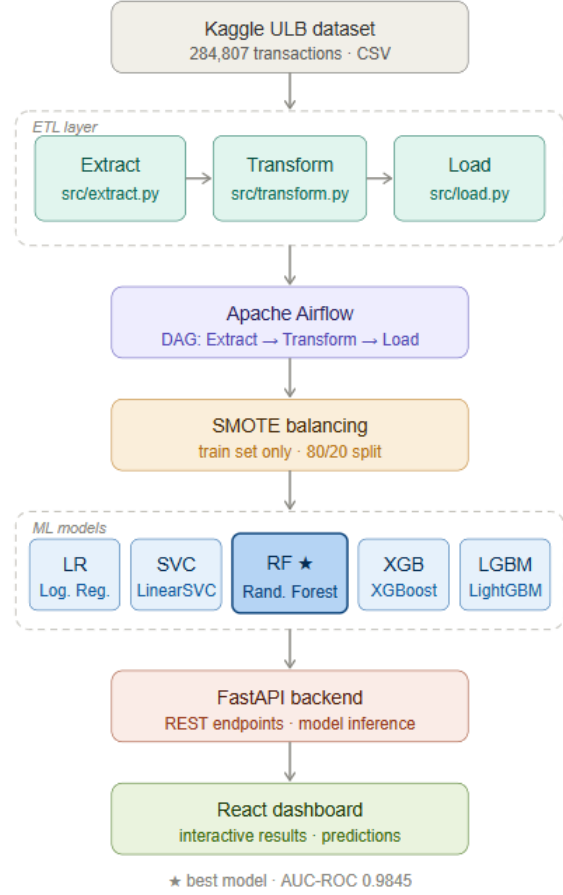


Figure 1. End-to-end system architecture. Raw transaction data flows through the ETL pipeline orchestrated by Airflow, is balanced with SMOTE, then passed to five ML models. Random Forest (★, AUC-ROC 0.9845) is served via FastAPI and visualized in a React dashboard.

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5. Dataset

5.1. Input and Output

The input is a transaction record $x \in R^{30}$ with three types of features: (1) 28 PCA-transformed anonymized features $V_1, \dots, V_{28}$ representing behavioral patterns, (2) Amount_Scaled, the transaction amount standardized using z-score normalization:

$$z = \frac{x - \mu}{\sigma}$$

and (3) `Hour`, the hour of day extracted from the raw `Time` field as:

$$\text{Hour} = \left\lfloor \frac{\text{Time}}{3600} \bmod 24 \right\rfloor$$

The output is a binary label $y \in \{0, 1\}$ where $y = 1$ indicates fraud.

5.2. Distribution, Size, and Missingness

The dataset contains 284,807 transactions. The class distribution is severely imbalanced: 284,315 legitimate (99.83%) and 492 fraudulent (0.17%). There are no missing values in any column. Most transaction amounts are small (under \$500 for both classes), but legitimate transactions have outliers up to \$25,691 while fraud transactions never exceed \$2,125 (Figure 2). Fraudulent transactions are more evenly spread across hours, with 2AM having the highest fraud rate at 1.713% (Figure 3).

5.3. Preprocessing

The preprocessing pipeline (`src/transform.py`) performs three steps: (1) engineer the `Hour` feature from the raw `Time` column, (2) scale the `Amount` column using `StandardScaler` and save the scaler artifact to `models/scaler.pkl` for consistent inference, and (3) drop the original `Time` and `Amount` columns. Median values of V_1 – V_{28} are saved to `models/v_medians.json` for use during single-transaction prediction.

5.4. Exploratory Data Analysis

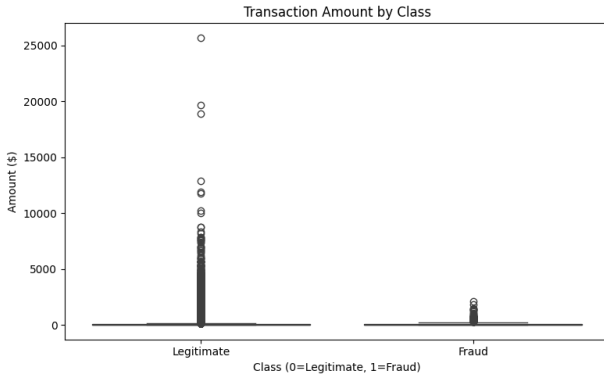


Figure 2. Transaction amount by class. Legitimate transactions have outliers up to \$25,691; fraud stays below \$2,125.

6. Methodology

6.1. Evaluation Metrics

Since the dataset is severely imbalanced, accuracy alone is misleading — a model predicting all transactions as le-

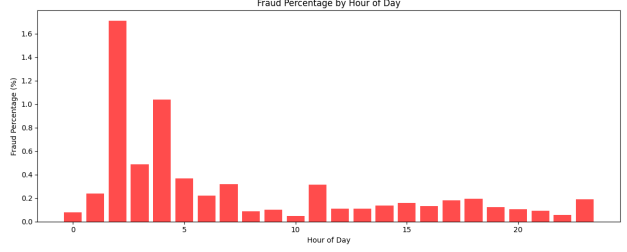


Figure 3. Fraud percentage by hour of day. 2AM has the highest fraud rate at 1.713%.

gitimate achieves 99.83% accuracy. Instead, the following metrics are used:

Precision measures how many flagged transactions are actually fraud:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall measures how many actual fraud cases the model catches:

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score balances precision and recall:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

AUC-ROC measures the model’s ability to distinguish between classes across all thresholds (0.5 = random, 1.0 = perfect).

6.2. Models

Baseline — Logistic Regression: A linear model that estimates fraud probability using the sigmoid function:

$$P(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}}$$

LinearSVC: A linear support vector classifier that finds the maximum-margin hyperplane by minimizing:

$$\frac{1}{2} \|w\|^2 + C \sum_i \max(0, 1 - y_i(w^T x_i + b))$$

Random Forest: An ensemble of decision trees trained on bootstrap samples. The final prediction is the majority vote across all trees, which reduces overfitting and handles class imbalance.

XGBoost: A gradient boosting method that builds trees sequentially, each correcting the residual errors of the previous trees using a regularized objective function.

LightGBM: A gradient boosting method using leaf-wise tree growth instead of level-wise, making it faster on large datasets.

6.3. Training and Validation

The data is split 80/20 into training (227,845 samples) and test (56,962 samples). SMOTE is applied only to the training set, balancing both classes to 227,451 samples each. The test set remains untouched to simulate real-world conditions. All five models are trained on the balanced training data and evaluated on the original imbalanced test set.

Table 1. Model comparison on the test set.

Model	Prec.	Recall	F1	AUC
Logistic Reg.	0.05	0.92	0.10	0.9715
LinearSVC	0.05	0.92	0.10	0.9742
Random Forest	0.86	0.83	0.84	0.9845
XGBoost	0.54	0.87	0.66	0.9739
LightGBM	0.10	0.91	0.19	0.9802

Random Forest achieves the best overall performance with the highest precision (0.86), F1-score (0.84), and AUC-ROC (0.9845).

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